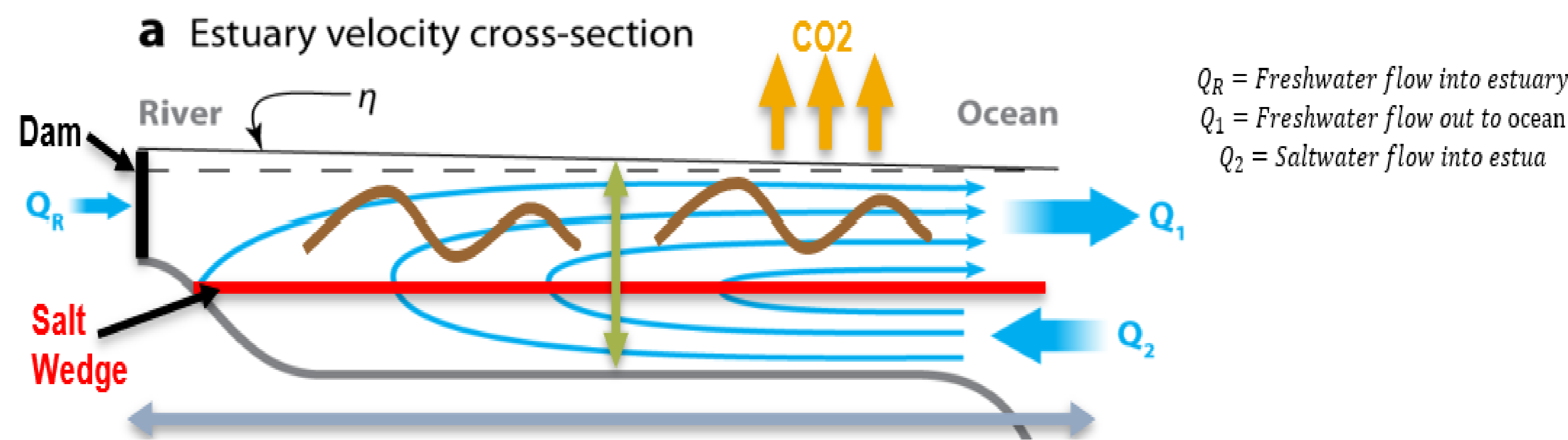


Background

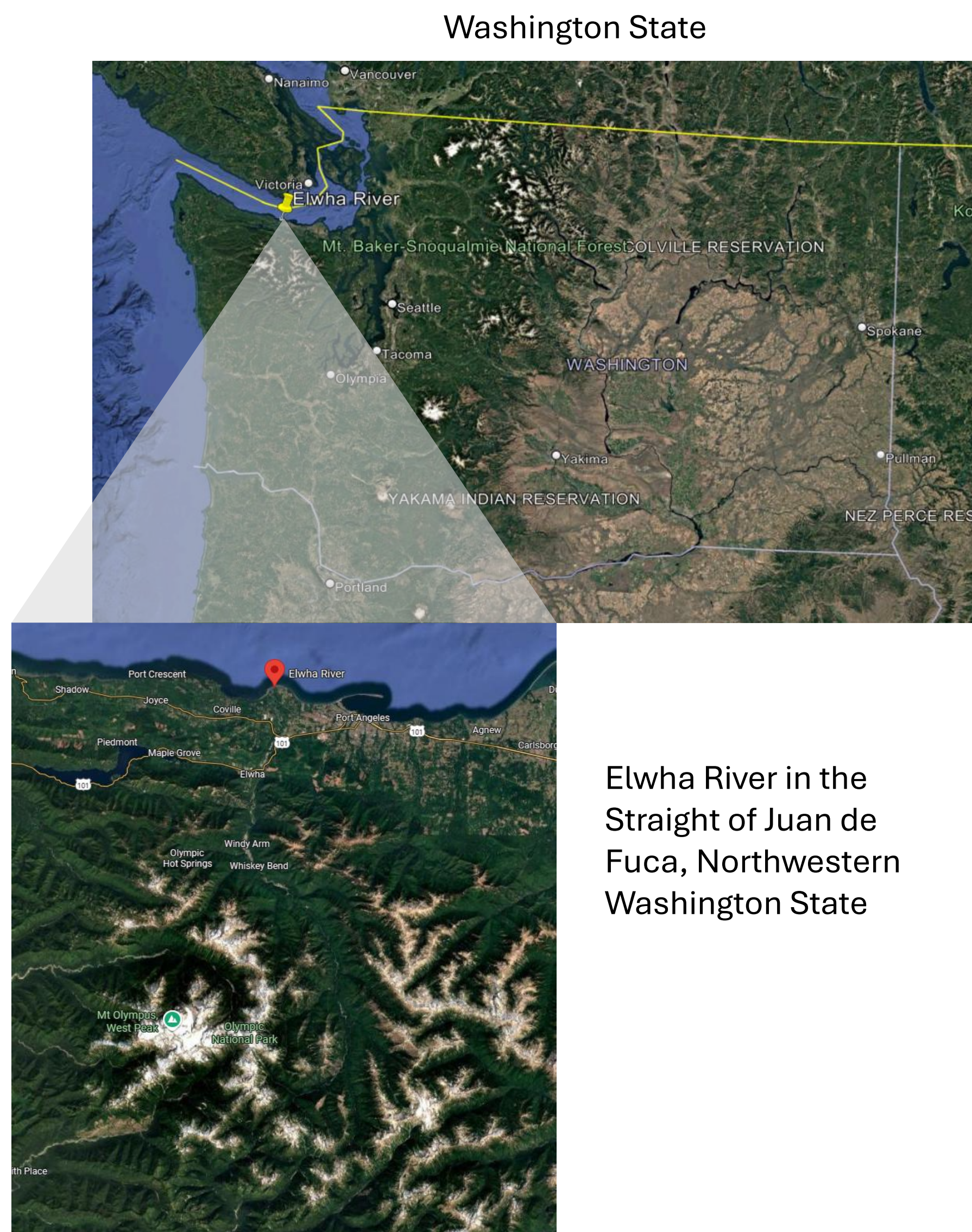
Estuaries are highly productive ecosystems where freshwater mixes with seawater, creating habitats that protect coastlines, filter pollutants, and support fisheries. Despite covering only 0.03% of the global ocean area, estuaries contribute about 17% of oceanic CO₂ flux, making them key players in the global carbon cycle. Dam construction disrupts sediment transport, alters salinity, and degrades water quality, while dam removal releases stored sediment and changes freshwater-saltwater balance. The Elwha River in Washington State, where two major dams were removed between 2011 and 2014, experienced a >100-fold increase in sediment deposition. While physical changes are well documented, the long-term biogeochemical impacts remain uncertain.



Elwha River Dam 8 km upstream



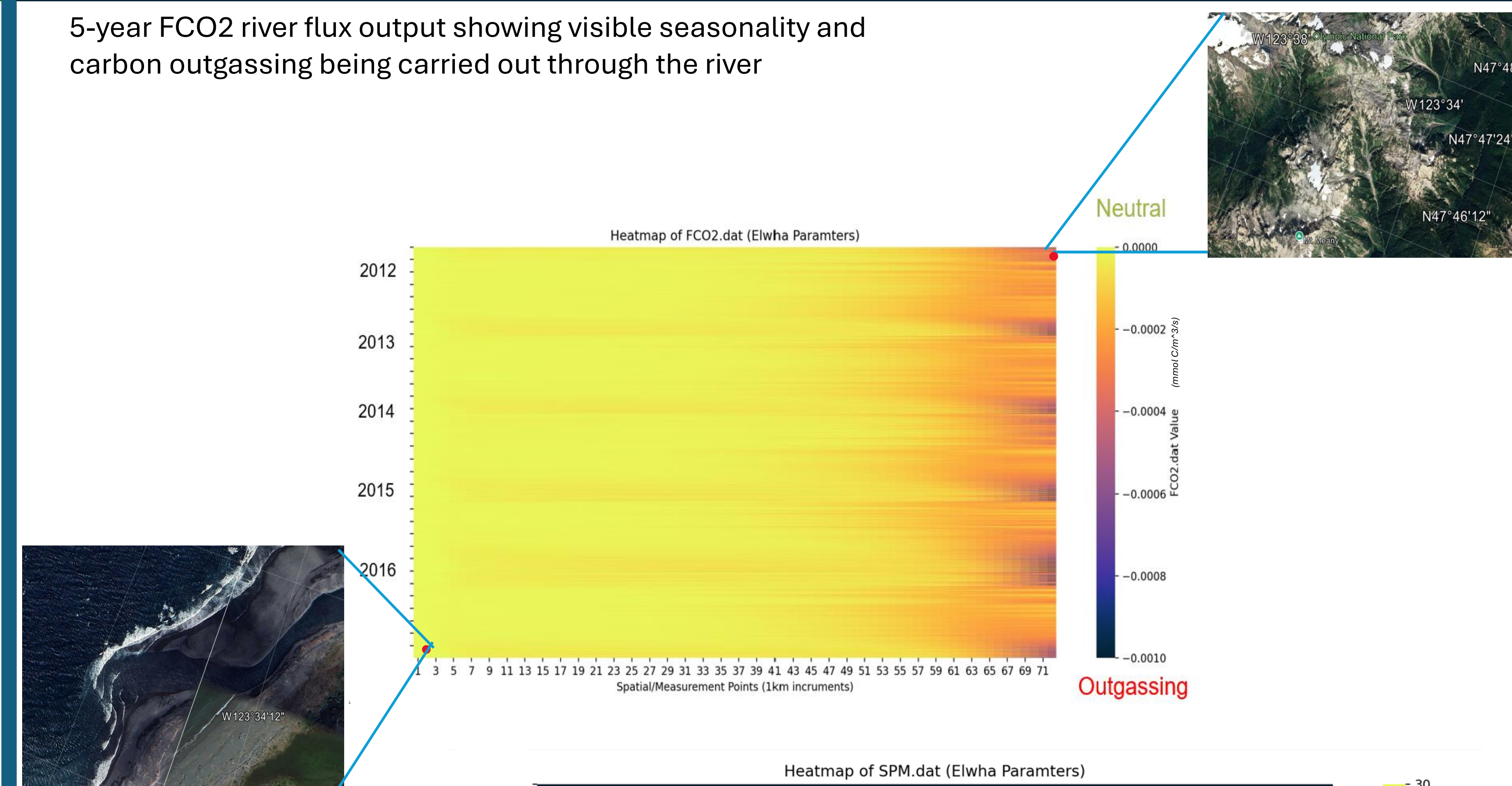
Glines Canyon Dam 22.5 km upstream



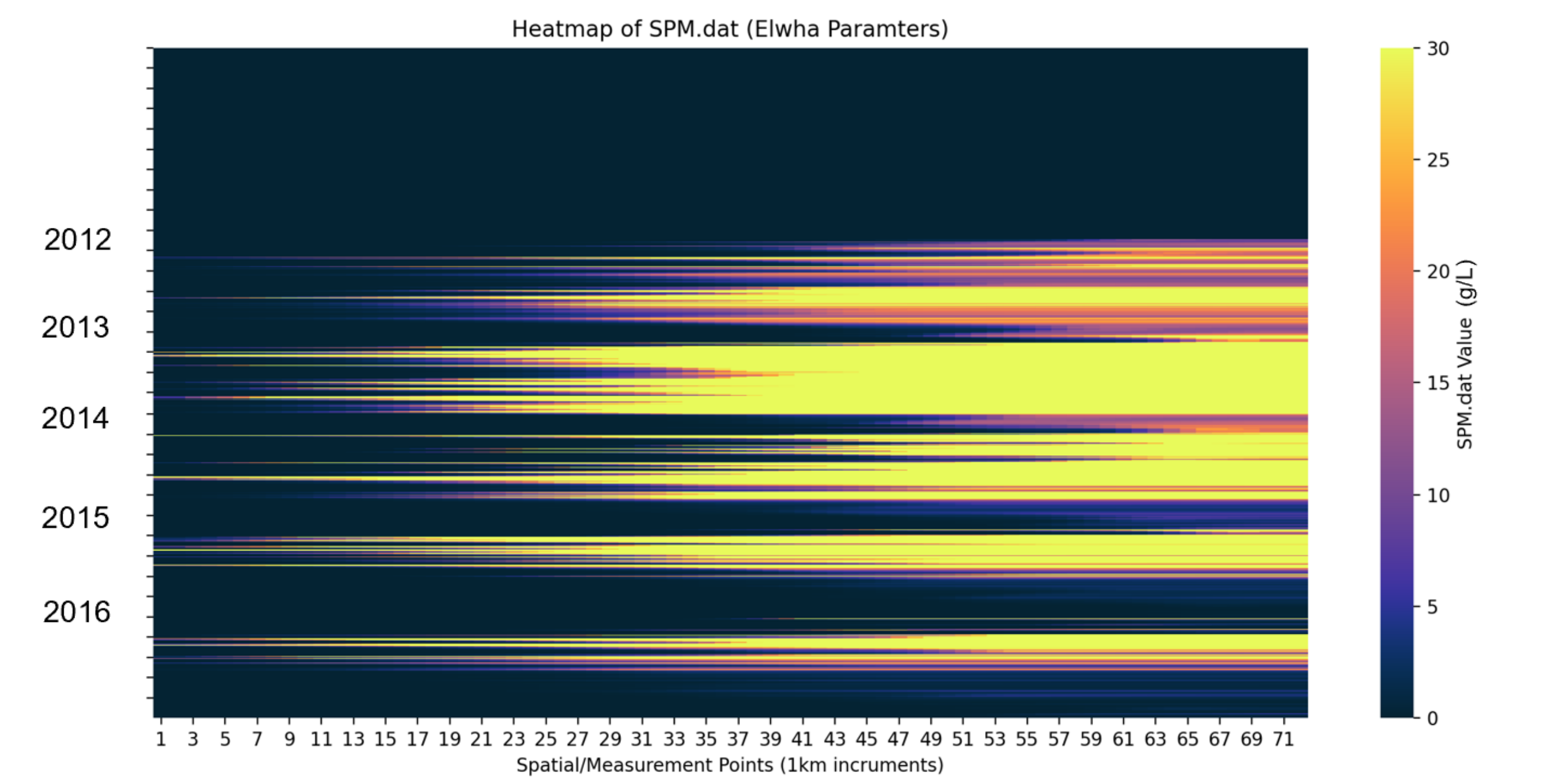
Elwha River in the Strait of Juan de Fuca, Northwestern Washington State

Results

5-year FCO₂ river flux output showing visible seasonality and carbon outgassing being carried out through the river

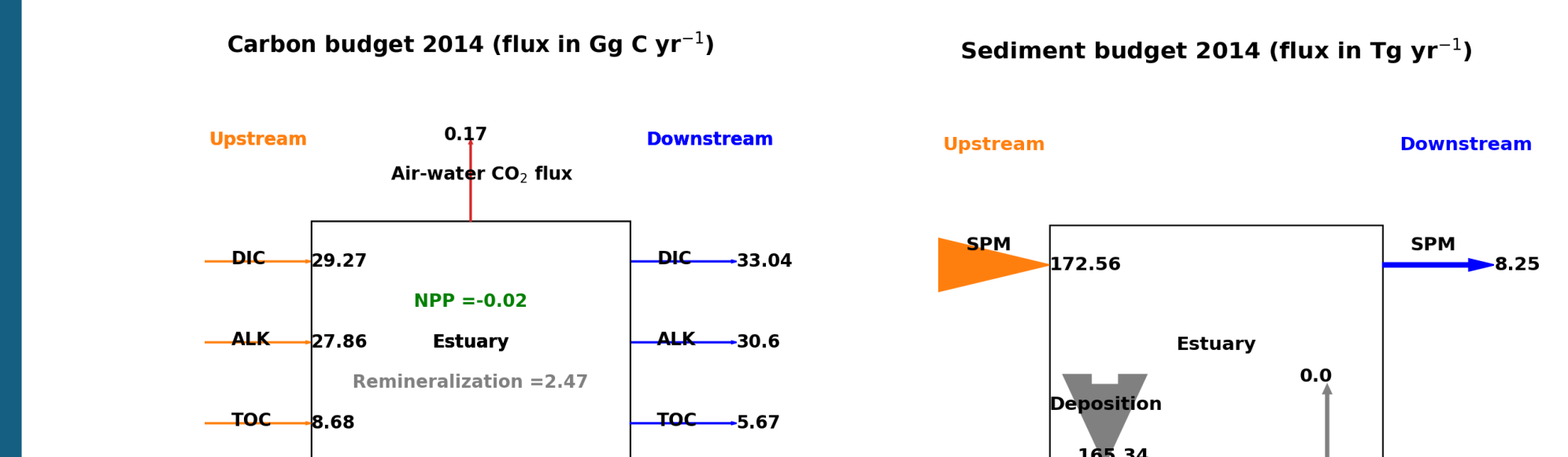
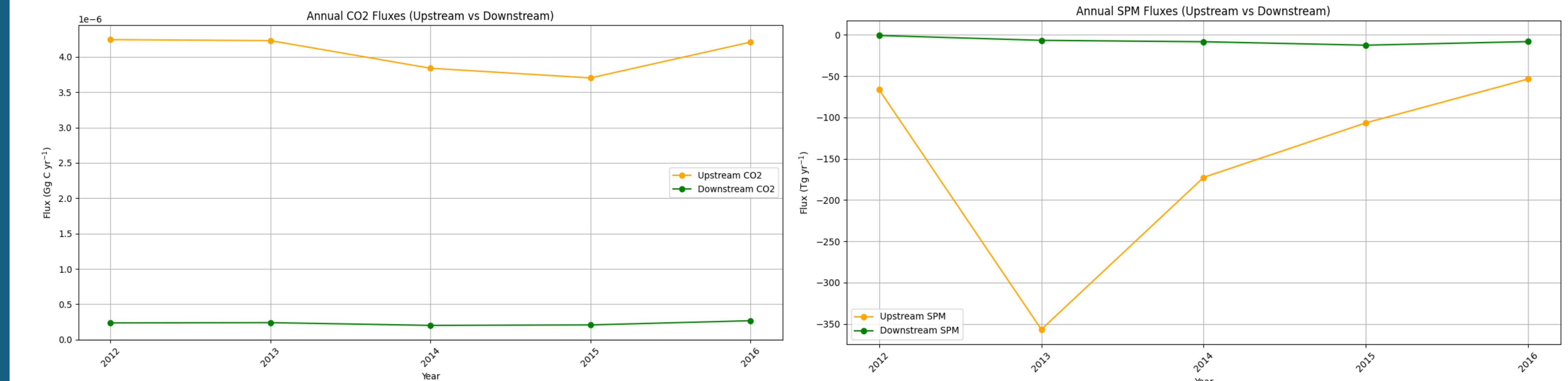
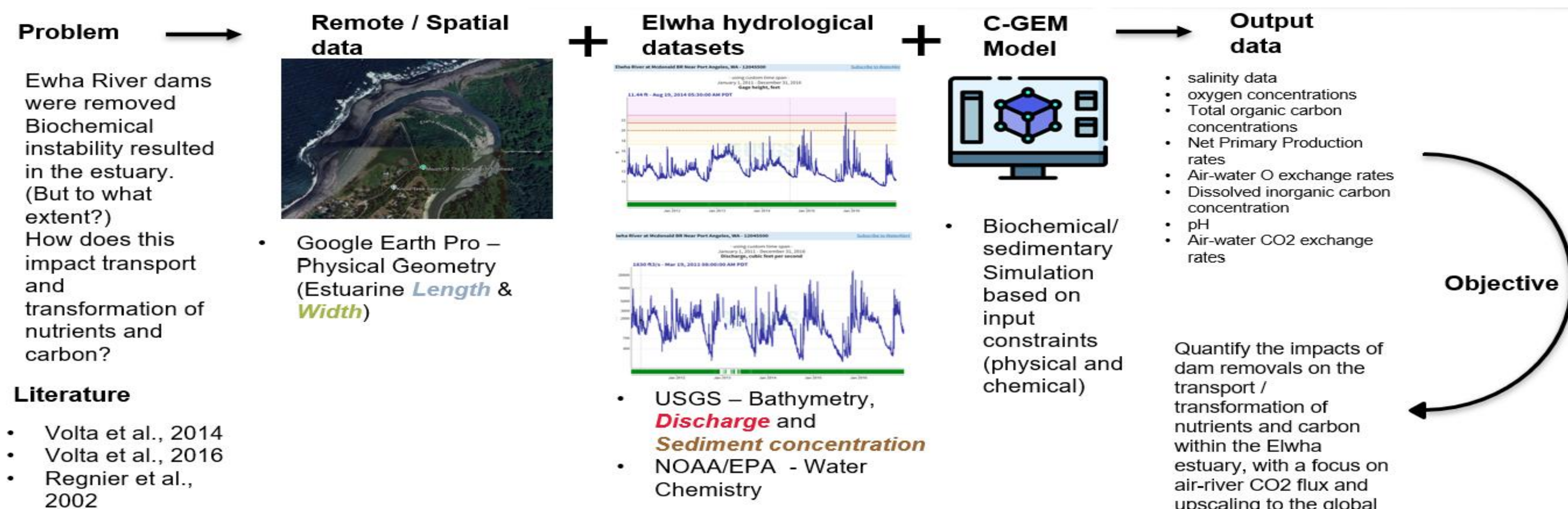


5-year sediment flux output showing visible highest peak of suspended sediment in the water which is confirmed by USGS surveys



Methodology

We configured the Carbon-Generic Estuary Model (C-GEM) to simulate the Elwha River estuary from 2011–2016 using site-specific environmental forcings. All hydrological, meteorological, atmospheric, and sediment datasets were linearly interpolated to match the model timestep. Hydrological inputs were derived from daily USGS discharge data, while meteorological variables, including wind speed and air temperature, were obtained from NOAA and NASA POWER datasets. Atmospheric pCO₂ forcing was replaced with an interpolated observational time series from NOAA records to capture seasonal and interannual variability. Suspended sediment inputs were based on USGS monitoring data following dam removal. The were analyzed using a custom Python pipeline to produce heatmaps, annual budgets, and spatial profiles of carbon flux and transport.



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